

IEAP-Series03-Assignment

Sarah Durand, Gaia Gassiole, Vidusha Thebuwana

2026-10-07

IEAP-Rstudio-Series03

Data wrangling and ANOVA

1 ANOVA: Analysis of variance

1.1 Think before coding

The scientific question is:

- Is the effect of difficulty different in patients with stroke compared to healthy?
 - How do you visualize this?
 - * ggplot
 - Is there a significant difference?
 - * ANOVA?
 - What sentence should we write to state this result in a report?

1.2 What library to answer the above questions?

Load library

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ez)
```

Registered S3 method overwritten by 'lme4':

```
method          from
na.action.merMod car
```

1.3 Data reorganization

Load data

```
data <- read.table(
  text = readLines("data/Results.txt"),
  header = TRUE,
  sep = "",
  stringsAsFactors = TRUE
)

head(data)
```

	GROUP	SUBJ	TRIAL	ORI	DIR	ID	REP	NbVelPeaks	MovementTime
1	ap	apS01	T01	W	outward	ID1	R1	2	1.23
2	ap	apS01	T01	W	inward	ID1	R1	2	1.26
3	ap	apS01	T01	NW	outward	ID1	R1	3	1.14
4	ap	apS01	T01	NW	inward	ID1	R1	2	0.90
5	ap	apS01	T01	N	outward	ID1	R1	3	1.11
6	ap	apS01	T01	N	inward	ID1	R1	3	0.81

We need to reorganize the data.

To do so, we count the number of data points for each combination of factors to check if there is missing data.

The following function looks for every unique combination of GROUP and ID in the dataset and tallies up the number of rows for each pair. It automatically generates a new column named n that contains these totals. we do that to check if the number of rows is the same for every combination.

```
data_counts <- data %>%  
  count(GROUP, ID)  
  
print(data_counts)
```

	GROUP	ID	n
1	ap	ID1	740
2	ap	ID2	740
3	ap	ID3	740
4	cp	ID1	1140
5	cp	ID2	1140
6	cp	ID3	1140
7	pp	ID1	760
8	pp	ID2	760
9	pp	ID3	400

For item 9 “pp ID3 400”, the number of rows does not match the rest of the group ‘PP’, so we need to delete ID3. The fact that some data are missing is explained by some stroke patients not being able to perform the test at ID3.

```
data %>%  
  filter(ID != "ID3") %>%  
  count(GROUP, ID)
```

	GROUP	ID	n
1	ap	ID1	740
2	ap	ID2	740
3	cp	ID1	1140
4	cp	ID2	1140
5	pp	ID1	760
6	pp	ID2	760

Now let's look at other columns such as REP.

```
data %>%  
  filter(ID != "ID3") %>%  
  count(REP)
```

	REP	n
1	R1	2040
2	R2	2040
3	R3	1200

The number of repetitions for R3 is half the number of R1 and R2. Therefore, we delete R3 data from our data frame.

```
data %>%  
  filter(ID != "ID3") %>%  
  filter(REP != "R3") %>%  
  count(GROUP, ID, REP)
```

	GROUP	ID	REP	n
1	ap	ID1	R1	260
2	ap	ID1	R2	260
3	ap	ID2	R1	260
4	ap	ID2	R2	260
5	cp	ID1	R1	380
6	cp	ID1	R2	380
7	cp	ID2	R1	380
8	cp	ID2	R2	380
9	pp	ID1	R1	380
10	pp	ID1	R2	380
11	pp	ID2	R1	380
12	pp	ID2	R2	380

We created a tibble 3×2 to look for the number of participants in each group.

```
data %>%  
  filter(ID != "ID3") %>%  
  filter(REP != "R3") %>%  
  distinct(GROUP, SUBJ) %>%  
  count(GROUP, name='count')
```

	GROUP	count
1	ap	13
2	cp	19
3	pp	19

Now we create a new variable to store the cleaned data.

```
data_clean <- data %>%
  filter(ID != "ID3") %>%
  filter(REP != "R3")
head(data_clean)
```

	GROUP	SUBJ	TRIAL	ORI	DIR	ID	REP	NbVelPeaks	MovementTime
1	ap	apS01	T01	W	outward	ID1	R1	2	1.23
2	ap	apS01	T01	W	inward	ID1	R1	2	1.26
3	ap	apS01	T01	NW	outward	ID1	R1	3	1.14
4	ap	apS01	T01	NW	inward	ID1	R1	2	0.90
5	ap	apS01	T01	N	outward	ID1	R1	3	1.11
6	ap	apS01	T01	N	inward	ID1	R1	3	0.81

Run the ANOVA

```
anova_results <- ezANOVA(
  data = data_clean,
  dv = .(MovementTime), # dependent variable
  wid = .(SUBJ), # to recognize each participant
  within = .(ID), # within-subject factor
  between = .(GROUP), # between-subject factor
  detailed = FALSE
)
```

Warning: You have removed one or more levels from variable "ID". Refactoring for ANOVA.

Warning: Data is unbalanced (unequal N per group). Make sure you specified a well-considered value for the type argument to ezANOVA().

Warning: Collapsing data to cell means. *IF* the requested effects are a subset of the full design, you must use the "within_full" argument, else results may be inaccurate.

```
anova_results$ANOVA
```

	Effect	DFn	DFd	F	p	p<.05	ges
2	GROUP	2	48	37.24408	1.720007e-10	*	0.57462110
3	ID	1	48	30.77618	1.221240e-06	*	0.07667596
4	GROUP:ID	2	48	11.58124	7.866206e-05	*	0.05882293

ANOVA p-value:

We format the p-value to avoid scientific notation and see the full number. That way, it's easier to interpret the results.

```
format(anova_results$ANOVA$p, scientific = FALSE, digits = 10)
```

```
[1] "0.000000000172000706" "0.000001221240178853" "0.000078662063814274"
```

Interpretation:

The ANOVA revealed significant main effects of GROUP and ID, as well as a significant GROUP $\times$ ID interaction, with all p-values below 0.01.

Detailed ANOVA results:

Effect of the GROUP: Movement Time is different between the three groups ($p < 0.01$).

Effect of ID: Movement Time is different between the two IDs ($p < 0.01$).

Interaction effect GROUP $\times$ ID: The effect of ID on Movement Time depends on the GROUP ($p < 0.01$). That is to say, Movement Time differs between the groups depending on the ID. For now, we do not know the direction of the effect, but we will see it in the graph below.

Plot Movement Time as a Function of Index of Difficulty

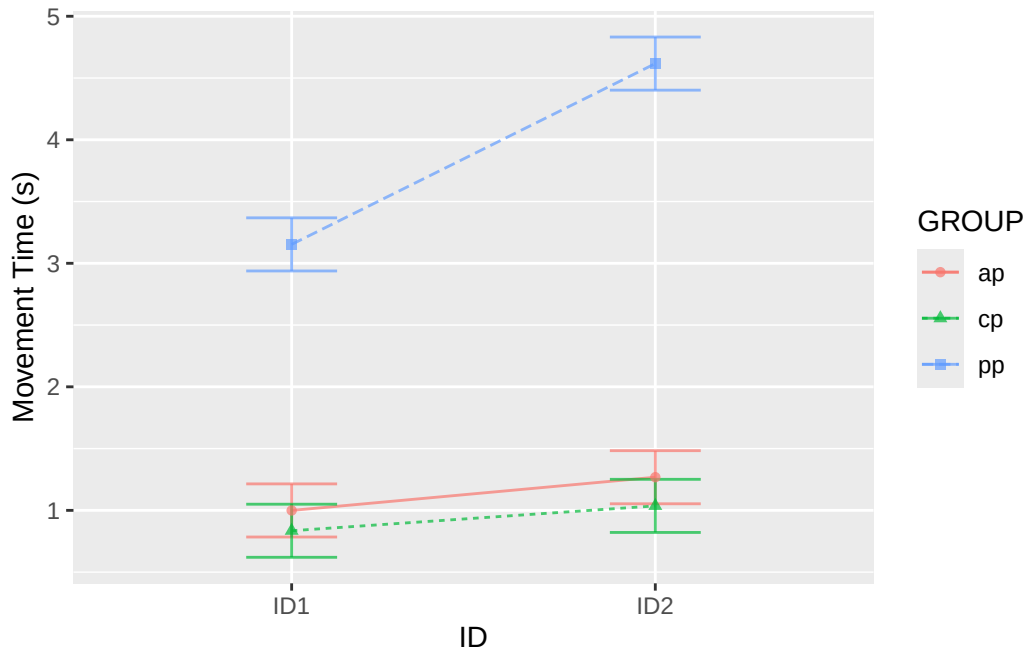
```
interaction_plot_reference <- ggplot(data_clean) +  
  aes(x = ID, y = MovementTime, color = GROUP) +  
  # lines: one line type per group  
  stat_summary(fun = mean, geom = "line",  
               aes(group = GROUP, linetype = GROUP), alpha = 0.7) +  
  # error bars: mean  $\pm$  0.215 s, same size for all groups  
  stat_summary(fun.data = function(x) data.frame(y = mean(x),  
                                                  ymin = mean(x) - 0.215,
```

```

                                ymax = mean(x) + 0.215),
    geom = "errorbar", width = 0.25, alpha = 0.7) +
  # points: one shape per group
  stat_summary(fun = mean, geom = "point",
    aes(shape = GROUP), size = 1.5, alpha = 0.7) +
  labs(x = "ID", y = "Movement Time (s)")

show(interaction_plot_reference)

```



Interpretation:

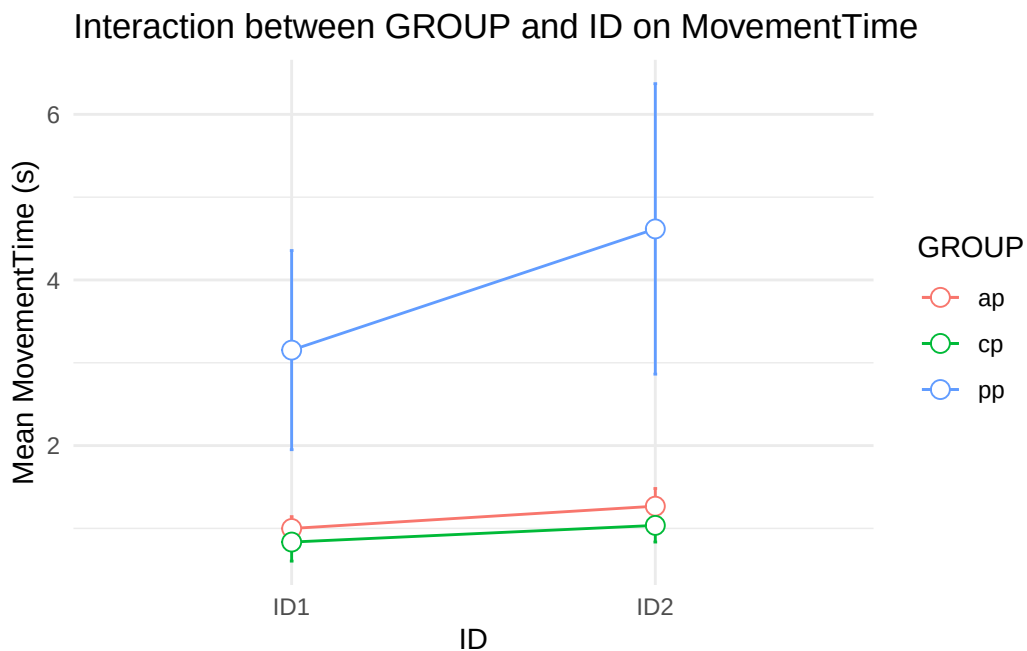
From the ANOVA analysis, we already know that there is a significant interaction effect between GROUP and ID. From the graph, we can see that Movement Time increases as the Index of Difficulty increases in all three groups. However, this increase is much stronger in the stroke patients group compared to the two other groups.

...Continuation of the section 1.3

```
library(ggplot2)
```

```
# Interaction between GROUP and ID, with error bars of +/- 0.5 SD around the mean
interaction_plot_Group_ID <- ggplot(data_clean) +
  aes(x = ID, y = MovementTime, color = GROUP) +
  stat_summary(fun = mean, geom = "line", aes(group = GROUP)) +
  stat_summary(fun.data = mean_sdl, fun.args = list(mult = .5),
    geom = "errorbar", width = 0.01) +
  stat_summary(fun = mean, geom = "point", shape = 21, fill = "white", size = 3) +
  labs(title = "Interaction between GROUP and ID on MovementTime",
    x = "ID",
    y = "Mean MovementTime (s)") +
  theme_minimal()

print(interaction_plot_Group_ID)
```



1.4 Run the linear regression by GROUP

First let's check the data types of each column. Linear regression requires a number not strings.

```
str(data_clean)
```

```
'data.frame':  4080 obs. of  9 variables:
 $ GROUP      : Factor w/ 3 levels "ap","cp","pp": 1 1 1 1 1 1 1 1 1 1 ...
 $ SUBJ       : Factor w/ 51 levels "apS01","apS02",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ TRIAL      : Factor w/ 2 levels "T01","T02": 1 1 1 1 1 1 1 1 1 1 ...
 $ ORI        : Factor w/ 5 levels "E","N","NE","NW",...: 5 5 4 4 2 2 3 3 1 1 ...
 $ DIR        : Factor w/ 2 levels "inward","outward": 2 1 2 1 2 1 2 1 2 1 ...
 $ ID         : Factor w/ 3 levels "ID1","ID2","ID3": 1 1 1 1 1 1 1 1 1 1 ...
 $ REP        : Factor w/ 3 levels "R1","R2","R3": 1 1 1 1 1 1 1 1 1 1 ...
 $ NbVelPeaks  : num  2 2 3 2 3 3 1 2 4 1 ...
 $ MovementTime: num  1.23 1.26 1.14 0.9 1.11 0.81 1.08 0.81 1.32 1.02 ...
```

Since ID is a categorical variable and it is better to run the actual numeric values before fitting to the linear model. Since ID1 was 3.8 and ID2 was 5.43 from the scientific paper, the data can be manipulated as follows:

```
data_reg <- data_clean %>%
  # Create a new column called 'ID_num', ID1 is 3.8, ID2 is 5.43
  mutate(ID_num = case_when(
    ID == "ID1" ~ 3.8,
    ID == "ID2" ~ 5.43
  ))
```

Now that data type is stored properly. Let's run the linear model.

```
models <- data_reg %>%
  # Split the data into separate group based on the 'GROUP' column
  group_by(GROUP) %>%
  # Loop through each pile (.x) and calculate a linear model (lm) predicting MovementTime from ID_num
  group_map(~ lm(MovementTime ~ ID_num, data = .x))

# Apply the actual group names (e.g., ap, cp, pp) as labels to the model list
names(models) <- levels(data_reg$GROUP)

# Loop that goes through each group name one by one (where 'g' represents the current name)
for (g in names(models)) {
  # Extract the math coefficients (intercept and slope) from the model belonging to group 'g'
  co <- coef(models[[g]])
  # Print the final equation as text, with 3 decimal places
  cat("Group", g, ": MT =", round(co[1], 3), "+", round(co[2], 3), "* ID\n")
}
```

Group ap : $MT = 0.372 + 0.165 * ID$
Group cp : $MT = 0.366 + 0.123 * ID$
Group pp : $MT = -0.259 + 0.898 * ID$

Interpretation of the data:

Group	Intercept (a)	Slope (b)	Meaning of the slope
ap	0.372 s	0.165 s/bit	each extra bit of difficulty adds about 0.17 s
cp	0.366 s	0.123 s/bit	each extra bit adds about 0.12 s
pp	-0.259 s	0.898 s/bit	each extra bit adds about 0.90 s

The pp group adds around 0.9s per unit increase in ID. This figure is much larger compared to the figures of ap(0.17s) and cp(0.12s).

1.5 Save a graph in a PDF file for publication

Construction of the graph

To reproduce a graph similar to Figure 3 of the article, we progressively construct the figure from the cleaned data.

First, we calculate the **mean Movement Time** and the **standard deviation** for each combination of group (ap, cp, and pp) and difficulty level (3.8 and 5.43 bits). This gives us six mean values, corresponding to the six points represented in the graph.

We use ID_num to represent the Index of Difficulty numerically on the x-axis.

The following table summarizes the variables used to construct the graph:

Variable	Role in the graph
GROUP	Defines the three groups: ap, cp, and pp
ID_num	Index of Difficulty on the x-axis: 3.8 and 5.43 bits
MovementTime	Movement Time measured for each trial
mean_MovementTime	Mean Movement Time represented by each point
sd_MovementTime	Standard deviation used to construct the error bars

We then use `ggplot2` to represent the six mean values. The Index of Difficulty is placed on the x-axis, the mean Movement Time on the y-axis, and a different colour is used for each group.

A linear regression line is then fitted separately for each group. The `geom_smooth()` function uses the `lm` method, which corresponds to R's linear model function. It automatically estimates the slope and intercept of the best-fitting linear regression line from the data.

Each regression line follows the equation:

$$y = a \times x + b$$

Letter	Mathematical name	In our graph
y	Measured variable	MT, the Movement Time
x	Explanatory variable	ID, the Index of Difficulty
a	Slope	Estimated by <code>lm()</code>
b	Intercept	Estimated by <code>lm()</code>

Therefore, for each group, the regression equation can be written as:

$$MT = b + a \times ID$$

The regression coefficients obtained in Section 1.4 are:

Group	Intercept (b)	Slope (a)
ap	0.372	0.165
cp	0.366	0.123
pp	-0.259	0.898

These values give the following regression equations:

Group	Regression equation
ap	$MT = 0.372 + 0.165 \times ID$
cp	$MT = 0.366 + 0.123 \times ID$
pp	$MT = -0.259 + 0.898 \times ID$

We add these equations to the graph using `annotate()`. They are positioned in the upper-left area of the graph, where there is enough free space.

Finally, the graph is modified to resemble Figure 3 of the article more closely. The axes and regression lines are extended, the two mean values of each group are connected with a solid line, error bars are added to represent variability, and the mean values are displayed as white circles with coloured outlines. The axis labels and overall theme are also adjusted.

```
# =====
# Step 1: Calculate the mean and standard deviation
# =====

# Compute the mean and standard deviation of Movement Time
# for each group and difficulty level.
group_means <- data_reg %>%
  group_by(GROUP, ID_num) %>%
  summarise(
    mean_MovementTime = mean(MovementTime),
    sd_MovementTime = sd(MovementTime),
    .groups = "drop"
  ) %>%
  # Reorder the groups (pp first) so that the colours match Figure 3 of the article
  mutate(GROUP = factor(GROUP, levels = c("pp", "ap", "cp")))

# Display the resulting table
# 6 rows = 3 groups x 2 difficulty levels
group_means
```

```
# A tibble: 6 x 4
  GROUP ID_num mean_MovementTime sd_MovementTime
  <fct> <dbl>         <dbl>         <dbl>
1 ap    3.8          0.999          0.285
2 ap    5.43        1.27           0.427
3 cp    3.8          0.835          0.460
4 cp    5.43        1.04           0.401
5 pp    3.8          3.15           2.40
6 pp    5.43        4.62           3.51
```

```
# =====
# Step 2: Create the basic graph
# =====

fitts_plot <- ggplot(
  data = group_means,
  aes(
    x = ID_num,
    y = mean_MovementTime,
    color = GROUP
  )
)
```

```

) +
  geom_point()

# =====
# Step 3: Add the linear regression lines
# =====

# lm automatically estimates the slope and intercept
# of the regression line for each group.
fitts_plot <- fitts_plot +
  geom_smooth(
    method = "lm",
    se = FALSE,
    linewidth = 0.5
  )

# =====
# Step 4: Add the regression equations
# =====

# The regression coefficients were obtained in Section 1.4.
# The equations are added manually using annotate().

fitts_plot <- fitts_plot +
  annotate(
    "text",
    x = 0,
    y = 6.5,
    label = "ap: MT = 0.372 + 0.165 * ID",
    hjust = 0,
    color = "#00BA38"
  ) +
  annotate(
    "text",
    x = 0,
    y = 6,
    label = "cp: MT = 0.366 + 0.123 * ID",
    hjust = 0,
    color = "#619CFF"
  ) +

```

```

annotate(
  "text",
  x = 0,
  y = 5.5,
  label = "pp: MT = -0.259 + 0.898 * ID",
  hjust = 0,
  color = "#F8766D"
)

# =====
# Step 5: Improve the graph to resemble Figure 3
# =====

# Extend the axes using invisible points.
fitts_plot <- fitts_plot +
  geom_blank(
    data = data.frame(
      x = c(-0.5, 6.5),
      y = c(-1, 7)
    ),
    aes(x = x, y = y),
    inherit.aes = FALSE
  ) +

# Extend the regression lines to the edges of the graph
# and display them as dashed lines.
geom_smooth(
  method = "lm",
  se = FALSE,
  fullrange = TRUE,
  linetype = "dashed",
  linewidth = 0.3
) +

# Connect the two mean values of each group.
geom_line() +

# Add error bars using half of the standard deviation.
geom_errorbar(
  aes(
    ymin = mean_MovementTime - 0.5 * sd_MovementTime,

```

```

    ymax = mean_MovementTime + 0.5 * sd_MovementTime
  ),
  width = 0
) +

# Display the mean values as white circles
# with coloured outlines.
geom_point(
  shape = 21,
  fill = "white",
  size = 4
)

# =====
# Step 6: Final formatting
# =====

fitts_plot <- fitts_plot +
  labs(
    x = "ID",
    y = "Mean MovementTime (s)"
  ) +
  theme_classic()

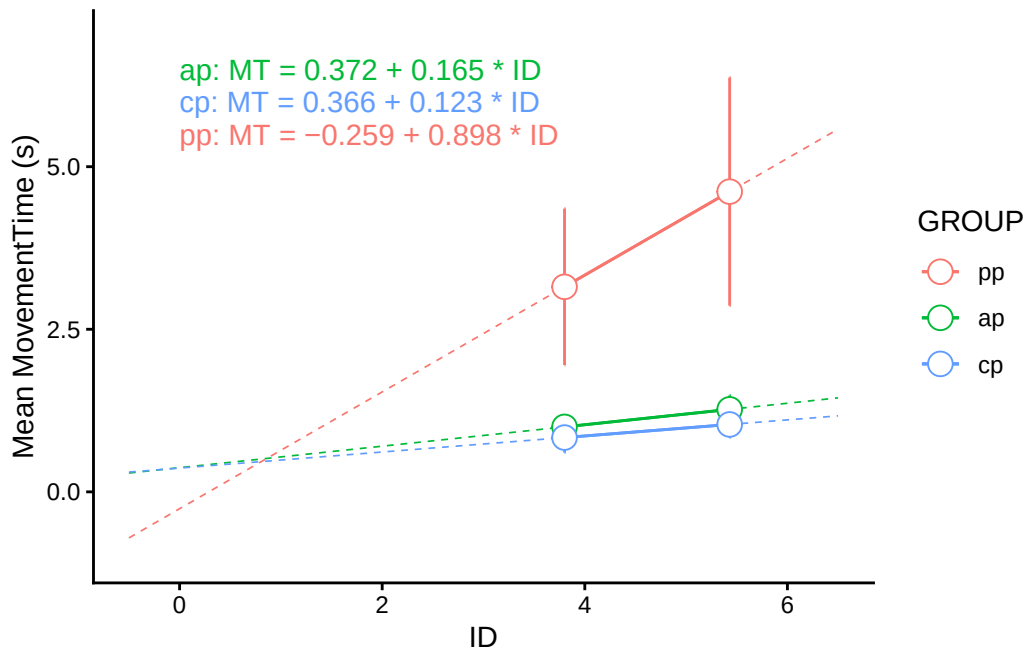
# Display the final graph
fitts_plot

```

```

`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'

```



Note on the code structure

The function `geom_smooth()` appears twice in our code:

- in **Step 3**, to show the basic regression line of each group;
- in **Step 5**, to extend these lines to the edges of the graph as dashed lines, as in Figure 3 of the article.

We chose to keep both on purpose, so that each step of our reasoning stays visible and easy to follow.

The code could be simplified by keeping only the dashed `geom_smooth()` of Step 5. The final graph would look the same, because the short solid line of Step 3 is hidden under the solid line added by `geom_line()`.

1.6 Export the graph as a PDF file for publication

```
ggsave(
  filename = here::here("Figure3_MT_vs_ID.pdf"),
  plot = fitts_plot,
  width = 8,
```

```
height =6,  
units = "in"  
)
```

```
`geom_smooth()` using formula = 'y ~ x'  
`geom_smooth()` using formula = 'y ~ x'
```